

Railway delays

An analysis of train delays on one of Italy's busiest intercity routes

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Introduction

This report analyses train delay patterns on the *Bologna–Milan* route, one of Italy's busiest intercity corridors. Using operational data from Trenitalia (December 2017 – March 2018), the analysis aims to identify the main temporal and spatial drivers of delays, and to build a predictive model estimating the probability of a significant delay (> 5 minutes).

The analysis is structured in three parts:

- (i) **Data preparation** — loading, cleaning, and variable engineering;
- (ii) **Exploratory analysis** — delay distributions, patterns by station, month, and day of the week;
- (iii) **Modelling** — linear regression for delay magnitude and logistic regression for delay probability.

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The raw dataset contains **32,806** observations across **6** variables: *date*, *train number*, *station*, *scheduled arrival*, *actual arrival*, and *delay in minutes*. The data covers the Bologna–Milan line and several connecting routes served by Trenitalia regional and intercity services.

A total of **16** records with delay values exceeding 180 minutes were removed as data entry errors — they represented less than 0.05% of observations and were clearly implausible¹. The final dataset contains **32,790** observations.

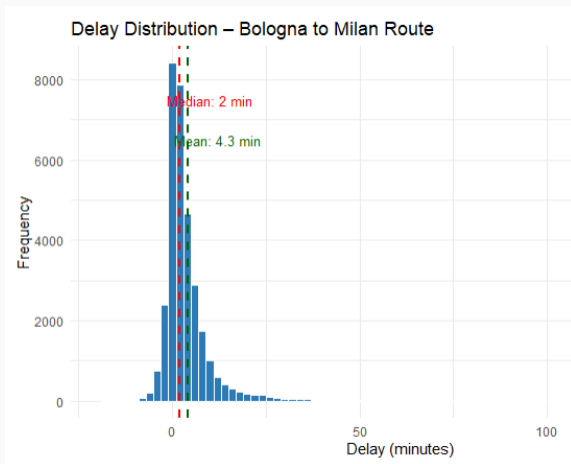
No missing values were found in the delay variable after parsing

¹The maximum raw value was approximately 1.06×10^{13} minutes

Exploratory Analysis

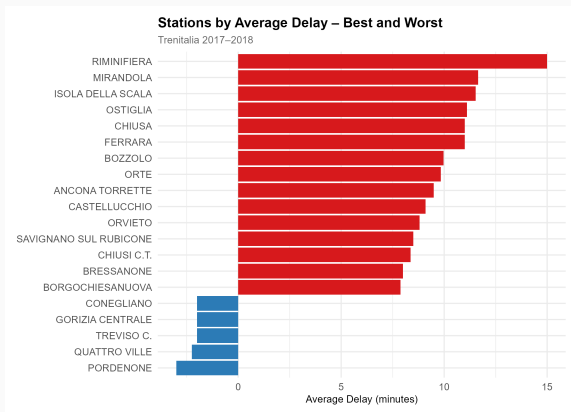
Delay Distribution

The distribution is right-skewed: most trains arrive on time or with minor delays, while a small proportion accumulates significant delays. The median delay is 2 minutes, while the mean is higher — driven upward by the long right tail. This gap between mean and median confirms that a minority of trains is responsible for the bulk of delay.



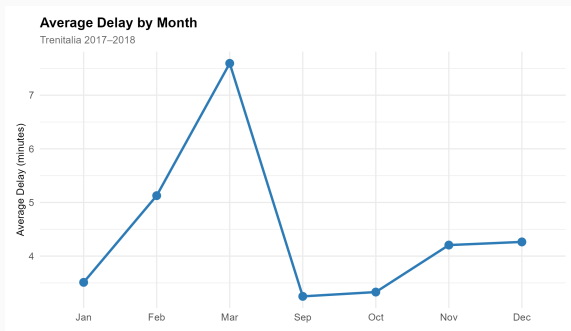
Delays by Station

Delay varies considerably across stops. High-delay stations tend to be intermediate stops, consistent with a delay propagation effect. Low-delay stations should be interpreted with caution, as they may reflect fewer observations rather than better punctuality.



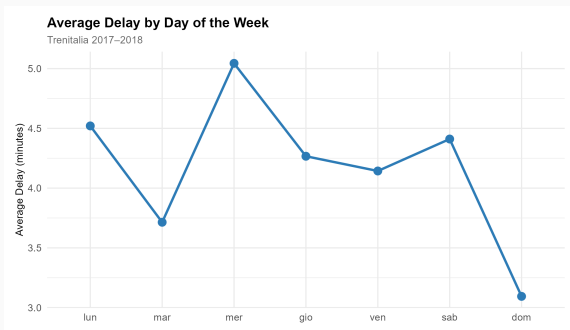
Delays by Month

The monthly pattern is not uniform. March is clearly an outlier, while the remaining months cluster between 2 and 5 minutes of average delay. This suggests that punctuality on this route is strongly influenced by episodic disruptions rather than a steady seasonal trend.



Delays by Day of the Week

The weekly pattern is coherent with expected traffic dynamics: weekdays show higher delays, with Wednesday as the peak, while Sunday benefits from reduced network load and fewer competing services.



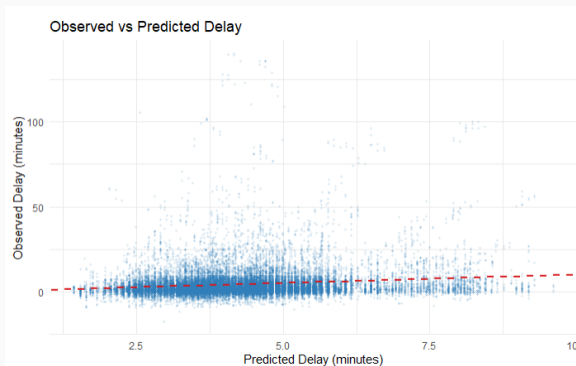
Modelling

A linear regression was fitted using month, day of the week, and hour of departure as predictors. The model achieves an R^2 of 0.023, meaning it explains approximately 2.3% of the variance in delay magnitude.

This low value is expected: the primary drivers of train delays — mechanical failures, weather events, and network disruptions — are not captured in this dataset. The model nonetheless confirms that temporal variables have a statistically significant effect, particularly month and day of the week.

Linear Regression

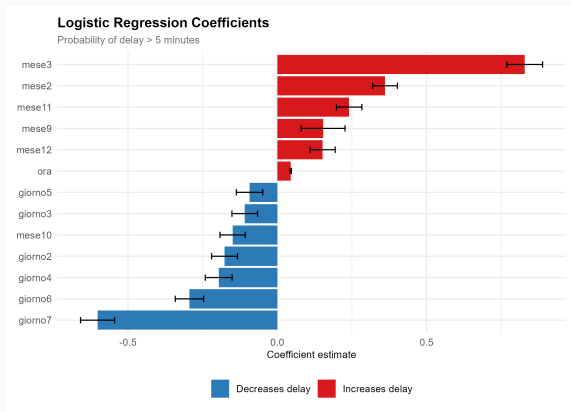
The scatter plot shows observed vs predicted delays. Points along the red dashed line would indicate a perfect fit. The wide dispersion confirms the low R^2 of 0.023: the model captures the general trend but lacks the external variables — weather, strikes, mechanical failures — needed for precise predictions.



A logistic regression was fitted to predict the probability of a significant delay, defined as exceeding 5 minutes. To address class imbalance — 74% of observations are non-delayed — the training set was resampled using the ROSE algorithm before fitting. The model was evaluated on a held-out test set (20% of the data). After resampling, balanced accuracy improves from 0.50 to 0.57, confirming that month, day of the week, and hour of departure carry genuine predictive signal. The AUC of 0.594 indicates limited but non-trivial discriminative power — a result consistent with the absence of operational variables such as rolling stock data or real-time network conditions.

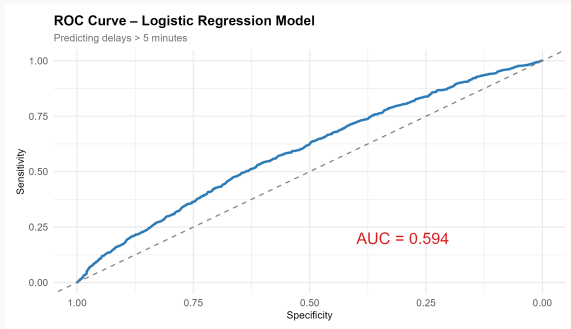
Logistic Regression

The coefficient plot confirms the patterns identified in the exploratory analysis. March has the largest positive effect on delay probability, while Sunday significantly reduces it. The hour of departure has a small but consistent positive effect, suggesting that delays accumulate throughout the day.



Logistic Regression

The model achieves an AUC of 0.594. While modest, this result is consistent with the limited set of predictors available. After ROSE resampling, the balanced accuracy on the test set improves from 0.50 (naive baseline) to 0.57, confirming that the temporal features carry genuine predictive signal.



Conclusions

The analysis of the Bologna–Milan route highlights several recurring patterns in train delays:

- (i) **March** is the worst month by a wide margin, suggesting the influence of seasonal disruptions such as strikes or adverse weather;
- (ii) **Wednesday** is the most delayed weekday; **Sunday** consistently shows the lowest delays, reflecting reduced network load;
- (iii) **Later departures** are associated with slightly higher delays, likely due to schedule propagation effects throughout the day;
- (iv) The predictive models confirm these patterns are statistically significant, though the low R^2 and AUC indicate that the primary drivers of delays remain unobserved.

Future work could integrate external data sources — weather records, strike calendars, rolling stock data — to substantially improve predictive performance and move toward an operationally useful delay forecasting tool.

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